Cyrille E. Mvomo

Graduate Student Researcher at Lake Lucerne Institute & McGill University

Lake Lucerne Institute AG
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He/Him, 26 yo

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(updated Aug 2025)
Citizen of France

Research Interests

Neuromechanics (Gait & Balance); Data Analytics; Wearable Technologies

Education

Montreal & Vitznau **Lake Lucerne Institute & McGill University**
09/2023 – 05/2027 Ph.D. in Biomechanics & Neurosciences (Project R.06 below)
Committee: Drs. Chris Awai *(Supervisor)*, Caroline Paquette *(Supervisor)*, Phil Dixon

Lausanne, CH **Swiss Federal Institute of Technology in Lausanne (EPFL)**
03/2023 – 08/2023 Visiting Graduate Student at the Institute of Bioengineering (Project R.05 below)
Referee: Drs. Alireza Manzoori *(Supervision)* and Mohamed Bouri

Paris, FR **University of Paris (now Paris Cité)**
09/2021 – 08/2023 M.S. Sciences, Technology, Health – AIRE Digital Sciences
Concentration: Data Science, AI, Research
Biomedical Specialization Modules at Paris Saclay University
Referee: Dr. Amodsen Chotia *(U. of Paris)*, Pr. Nathalie Lassau *(Paris Saclay U.)*

03/2018 – 08/2021 B.S. Grade in Allied Health Sciences (Project R.02 below)
University grade for the "Diplôme d'état" in Podiatry
Entry via Pre-med contest (PACES 2016-2018)
Referee: Pr. Yves Matillon

Recognitions

Labels:
Merit-based
Need-based

Awards

2024	Travel Award, Parkinson Canada (PDCA)	~580 CHF
2024	Elevator Pitch Competition People's Choice Award, CRIR Montréal	~30 CHF
2024	Travel Award, Canadian Institute of Health Research (CIHR)	~990 CHF
2023	M.S. Summa Cum Laude, University of Paris	

2023	M.S. Top 3 Ranked, University of Paris	

Fellowships

2024 – 2025	Ph.D. Recruitment Fellowship, Quebec BioImaging Network	~5800 CHF
2023 – 2026	Graduate Excellence Fellowship, McGill University	~35000 CHF
2023 – 2024	Anti-Black Racism Initiative Graduate Excellence Fellowship, McGill U.	~5800 CHF
2023	SMARTS-UP Scholarship, University of Paris	~4510 CHF
2022	Visiting Scholarship for International Students, cereneo Foundation	5000 CHF
2017 – 2023	Need-based Scholarship, Paris CROUS and IDF region	~3980 CHF

Honours

2025	Invited for Undergraduate Thesis Defense Jury, AFREP, University of Paris
2025 – Now	Student Member, Institute of Electrical and Electronics Engineers (IEEE)
2025	Guest judge for the 3-minute thesis competition, McGill University
2024 – Now	Student Member, Movement Disorders Society (MDS)
2024 – Now	Student Member of the PIER Committee (Innovative Practice In Research Ethics), CRIR
2023 – Now	Student Member, Provincial Adaptation-Readaptation Research Network (REPAR)
2023 – Now	Student Member, Center for Interdisciplinary Research in Rehabilitation Montréal (CRIR)
2023 – Now	Student Member, CRBLM, Faculty of Medicine & Health Sciences, McGill University
2023 – Now	Student Member, Quebec BioImaging Network (QBIN)
2023 – Now	Member of the Research Committee of the Rehabilitation Department, University of Paris
2023	Guest student member of the admission committee for the AIRE master's program, U. Paris

Professional Experience

2024 – Now **Office of Science Education, Faculty of Sciences, McGill University**

Montreal, CA Graduate Student Science Education Fellow (Workshops and Talks below)

Referee: Marcy Slapcoff

My main role is to develop content and teach (in collaboration with cognitive neuroscientists and education developers) the basic principles of neuroeducation to undergraduate students and instructors (teaching assistants and professors) in the Faculty of Sciences. I previously managed a collaborative learning space dedicated to first year science undergrad students.

2024 – Now **Department of Kinesiology and Physical Education, McGill University**

Montreal, CA Teaching Assistant (Courses below)

Referee: Dr. Caroline Paquette (Instructor)

My main role is to assist Dr Paquette in her course, in particular by addressing students' incomprehension, mentoring them during tutorials, as well as supervising and marking both written and oral exams.

2024 – Now **Department of Rehabilitation, Faculty of Health, AFREP, University of Paris**

Paris, FR Lecturer (Courses below)

Referee: Pr. Yves Matillon

My main role is to develop and deliver 20 hours of online courses on the basics of research methodology, with one part in the fall and the other in the winter.

2023 – Now **McGill HBCL & LLUI DART Laboratories**

Montreal & Vitznau Graduate Student Researcher (Publications and Presentations below)

Referee: Drs. Chris Awai (Supervisor), Caroline Paquette (Supervisor)

My research combines neuroimaging, data analytics and wearable sensing to rethink the development and validation of digital biomarkers to monitor motor symptoms in Parkinson's disease. In addition, I supervise undergraduate assistants and help my peers with their research projects in movement neuroscience.

2023 **EPFL REHAssist Group (BIOROB & TNE)**

Lausanne, CH Research Intern (Project R.05 below)

Referee: Dr. Alireza Manzoori

My main role was to evaluate the biomechanical parameters that explain why one mode of hip exoskeleton assistance strategy is more effective than another in reducing the metabolic cost of walking, by collecting and analyzing multimodal data (inertial motion capture, exoskeleton encoder, FSR, force plate, etc.).

2022 **INSERM UMR U1284 (CRI-LPI MakerLab)**

Paris, FR Research Intern (Project R.04 below)

Referee: Dr. Kevin Lhoste

My main role was to prototype a mobile application for fall-risk assessment in older adults, based on simple balance tests using accelerometer signals and machine learning.

2022 **LLUI DART Laboratory**

Vitznau, CH Research Intern (Project R.03, Paper J.01, Poster P.01 below)

Referee: Dr. Chris Awai

My main role was to collaboratively collect, analyze, and present data to clinical partners and international audiences from a project integrating motion capture (CAREN, Xsens), virtual reality, and wearable technologies to develop a novel visual–sensory contribution to balance assessment.

2021 – 2022 **Private Practice (20%)**

Évry, FR Podiatrist

Referee: Prisca Lonn *(Head – Simone Veil Community Health Center)*

Sport-oriented practice on Sunday mornings.

2021 **UTC Biomechanics & BioEngineering Laboratory** (Project R.01 below)

Compiègne, FR Undergraduate Research Initiation Intern

Referee: Dr. Khalil Ben Mansour

Initiation to motion capture and pressure sensing.

2021 **Motion Analysis Lab – Garches Foundation**

Garches, FR Undergraduate Research Initiation Intern

Referee: Dr. Anthony Supiot

Initiation to motion capture in neurorehabilitation.

2020 **IMSS Stade Français Paris Rugby Club**
Paris, FR Intern in Sports Podiatry
Referee: Christophe Blanc
Experience in Elite Rugby.

2018 – 2021 **Assistance Public Hopitaux de Paris (AP-HP)**
Paris, FR Intern in Podiatry
Referee: Pr Yves Matillon
Clinical training within Europe largest university hospital center.

Publications & Presentations

Labels:
Travel Award
International Conference
Peer-reviewed
People's Choice Award
* *Equal Contribution*

Journal Papers (J)

J.04 **Cyrille E. Mvomo**, Jordan Bedime, Dahlia Leibovich, Clara Guedes, Alexandra Potvin-Desrochers, Philippe C. Dixon, Chris Awai Easthope, Caroline Paquette. Sensitive Mobility Features in Parkinson's Disease: From Behavioral to Mechanistic Insights. *In preparation for Movement Disorders (2025)*

J.03 **Cyrille E. Mvomo***, Jordan Bedime*, Dorelle C. Hinton, Tomoha Ogawa, Alexander Thiel, Jean-Paul Soucy, Alexandra Potvin-Desrochers, Caroline Paquette. The Brain Network that Preserves Gait Adaptation in Parkinson's Disease. *In preparation for Brain, A Journal of Neurology (2025)*

J.02 Connor Delaney, Valerie Gibson, **Cyrille E. Mvomo**, Chris Awai Easthope, Parag Chitnis, Quentin Sanders. Emerging Technologies and Future Directions in the Detection, Prediction, and Management of Freezing of Gait in Parkinson's Disease. *In preparation for the Journal of NeuroEngineering and Rehabilitation (2025)*

J.01 Saskia Neumann, **Cyrille E. Mvomo**, Deepak Ravi, Friederike Schulte, Lorenz Assländer* Chris Easthope Awai*. Visual Dependence for Postural Control is Increased in Older Adults. *Submitted to Nature Communications Medicine (2025)*

Conference Papers (C)

C.01 **Cyrille E. Mvomo**, Jordan Bedime, Sara Perfetto, Dahlia Leibovich, Clara Guedes, Alexandra Potvin-Desrochers, Philippe C. Dixon, Chris Awai Easthope, Caroline Paquette. Monitoring Parkinson's Disease In-the-Wild. *Accepted for publication in IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25) (2025)*

Published Conference Abstract (A)

A.02 **Cyrille E. Mvomo**, Jordan Bedime, Isabella Sierra, Sara Perfetto, Henri Lajeunesse, Alexandra Potvin-Desrochers, Chris A. Easthope, Caroline Paquette. Model-Based Gait Outcomes to Track Parkinson's Disease Progression ? *In Movement Disorders (2024)* CIHR and PDCA Travel Awards

A.01 Isabella Sierra, Sara Perfetto, Alexandra Potvin-Desrochers, **Cyrille E. Mvomo**, Jordan Bedime, Henri Lajeunesse, Frederike Parent-L'Ecuyer, Caroline Paquette. Influence of Fatigue in Parkinson's Disease on Variability of Gait Measures. *In Movement Disorders (2024)*

Conference Posters (P)

P.12 **Cyrille E. Mvomo**, Jordan Bedime, Sara Perfetto, Dahlia Leibovich, Clara Guedes, Alexandra Potvin-Desrochers, Philippe C. Dixon, Chris Awai Easthope, Caroline Paquette. Monitoring Parkinson's Disease In-the-Wild. *Accepted for presentation at IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25) (2025)*

P.11 Jordan Bedime, **Cyrille E. Mvomo**, Sara Perfetto, Isabella Sierra, Henri Lajeunesse, Frédérike Parent-L'Ecuyer, Tomoha Ogawa, Jae Young Park, Madeleine Sharp, Marie-Hélène Pennestri, Alexandra Potvin-Desrochers, Bruno G. G. da Costa, Audrey Parent, Caroline Paquette. Physical activity and sedentary behavior monitoring in people with Parkinson's disease and fatigue. *At the International Society of Posture & Gait Research (ISPGR) - World Congress (2025)*

P.10 Jordan Bedime, **Cyrille E. Mvomo**, Sara Perfetto, Isabella Sierra, Henri Lajeunesse, Jean-Paul Soucy, Alexandra Potvin-Desrochers, Madeleine Sharp, Annie Rochette, Audrey Parent, Caroline Paquette. Fatigue in Parkinson's disease is linked to enhanced activity of the fornix and dysregulation of the dorsal attention and central executive networks during complex walking: An [18F]-FDG PET study. *At the Annual Canadian Movement Disorders Meeting (2024)*

P.09 **Cyrille E. Mvomo**, Isabella Sierra, Jordan Bedime, Sara Perfetto, Henri Lajeunesse, Alexandra Potvin-Desrochers, Chris A. Easthope, Caroline Paquette. Model-Based Gait Outcomes to Track Parkinson's Disease Progression ? *At the International Parkinson and Movement Disorder Society (MDS) Congress (2024)* CIHR and PDCA Travel Awards

P.08 Isabella Sierra, Sara Perfetto, Alexandra Potvin-Desrochers, **Cyrille E. Mvomo**, Jordan Bedime, Henri Lajeunesse, Frederike Parent-L'Ecuyer, Caroline Paquette. Influence of Fatigue in Parkinson's Disease on Variability of Gait Measures. *At the International Parkinson and Movement Disorder Society (MDS) Congress (2024)*

P.07 Sara Perfetto, Alexandra Potvin-Desrochers, Isabella Sierra, **Cyrille E. Mvomo**, Kevin Shoemaker, Mallar Chakravarty, Caroline Paquette. Long-term, high-level aerobic training and the age-related decline in cortical thickness. *At the Society for Neurosciences (SfN) World Congress (2024)*

P.06 Jordan Bedime, **Cyrille E. Mvomo**, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Madeleine Sharp, Rochette Annie, Jean-Paul Soucy, Audrey Parent, Caroline Paquette. Brain metabolism in Parkinson's disease with and without Fatigue during complex walking: An [18F]-FDG PET study. *At the Quebec Bioimaging Network (QBIN) Scientific Day (2024)*

P.05 **Cyrille E. Mvomo**, Jordan Bedime, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Caroline Paquette. A gait model that reflects measures of symptoms severity in Parkinson's disease (Part 2). *At the 5th Quebec Congress in Adaptation-Rehabilitation Research (2024)*

P.04 **Cyrille E. Mvomo**, Jordan Bedime, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Caroline Paquette. A gait model that reflects measures of symptoms severity in Parkinson's disease (Part 1). *At McGill Minds in Motion Conference (2024)*

P.03 Jordan Bedime, **Cyrille E. Mvomo**, Sara Perfetto, Henri Lajeunesse, Isabella Sierra, Alexandra Potvin-Desrochers, Caroline Paquette. The influence of cognitive load on locomotor adaptation to split-belt treadmill in Parkinson's disease with freezing of gait. *At McGill Minds in Motion Conference (2024)*

P.02 Saskia Neumann, **Cyrille E. Mvomo**, Deepak Ravi, Friederike Schulte, Lorenz Assländer, Chris A. Easthope. Relative importance of the visual system for balance control is reflected in the continuous margins of stability while walking. *At the XXIX Congress of International Society of Biomechanics (ISB) (2023)*

P.01 **Cyrille E. Mvomo**, Deepak Ravi, Friederike Schulte, Lorenz Assländer, Chris A. Easthope. Does locomotor perturbation training induce a change in relative sensory contribution to balance? A pilot study. *At RehabWeek (International NeuroRehabilitation Symposium) (2022)*

Oral Presentations (T)

T.02 **Cyrille E. Mvomo.** What if walking could help track the progression of Parkinson's disease? *TED-like talk at NeuroLingo, Montreal, CA (Fall 2024) – Knowledge Translation*

T.01 **Cyrille E. Mvomo.** What if walking could help track the progression of Parkinson's disease? *3-minutes thesis at CRIR 2024 Student Colloquium, Montreal, CA (Spring 2024) – Knowledge Translation People's Choice Award*

Research Projects

R.06 **Cyrille E. Mvomo.** (On going) *From Behavioral to Mechanistic Insights: Data-driven Exploration of Brain Locomotor Networks for Developing Digital Biomarkers of Motor Decline in Early Parkinson's Disease*. LLUI & McGill Ph.D. Thesis, Montreal & Vitznau, CH & CA

R.05 **Cyrille E. Mvomo.** (2023) *Investigation of Biomechanical Mechanisms Underlying the Efficacy of an Active Hip Exoskeleton Using PCA.* University of Paris & EPFL Final M.S. project, Paris & Lausanne, FR & CH

R.04 **Cyrille E. Mvomo.** (2023) *Identifying Falls-Prone Older Adults Using Smartphone Sensing : A Machine Learning Approach.* University of Paris & LPI M.S. semester project, Paris, FR

R.03 **Cyrille E. Mvomo.** (2022) *Does locomotor perturbation training induce a change in relative sensory contribution to balance? A pilot study.* University of Paris & LLUI M.S. semester project, Paris & Vitznau, FR & CH

R.02 **Cyrille E. Mvomo.** (2021) *Effect of foot orthoses on static balance in young adults.* University of Paris B.S. Thesis, Paris, FR

R.01 **Cyrille E. Mvomo.** (2021) *Effect of walking speed on peak plantar pressure.* Compiègne University of Technology Internship Thesis, Compiègne, FR

Service

Open-Source Contribution

2023 **MakerRehab**, CRI (LPI), University of Paris
Prototype of an mHealth technology to predict fall risk from smartphone accelerometers.

2022 **Into the Darkness**, CRI (LPI), University of Paris
Prototype of a virtual reality experience designed to elicit empathy in people affected by depression.

2022 **Blind Check**, CRI (LPI), University of Paris
Prototype of a robotic guide dog for people with visual impairments.

Social Mobilization

2024 **Patient Engagement Workshop**, Parkinson Canada
Participation in a workshop aimed at improving interaction between researchers, patients, clinicians and industrial partners to enhance patient engagement in Parkinson's research.

2022 **Hackathon**, HackaHealth Zürich
Participated in hackathon to develop solutions for people with disabilities.

2017 **Fundraising Events**, Or Bleu
Participation in soccer tournaments to finance the construction of water wells in Mali.

Organizing Committee

Fall 2025 **SciLearn Orientation Week Workshop**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Fall 2025 **SciLearn TA Training Workshop "How Students Learn"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing teaching assistants in Chemistry with basic knowledge of neuroeducation.

Winter 2025 **SciLearn Orientation Workshop "Prep for Finals"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2025 **SciLearn Orientation Workshop "MidTerm Bounce-Back"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2025 **Undergraduate Science Showcase 2025**, McGill University
Co-organizer of a conference aimed at providing undergrad science students the opportunity to present their research.

Fall 2024 **SciLearn Orientation Workshop "Prep for Finals"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Fall 2024 **SciLearn Orientation Workshop "MidTerm Bounce-Back"**, McGill University
Co-facilitator and co-organizer of a workshop aimed at providing undergrad science students with basic knowledge of neuroeducation.

Winter 2023 **AIRE Master Seminar 2023 "AI & Medicine"**, CRI (LPI), University of Paris
Co-organizer of a workshop on AI in medicine.

Peer Reviewing

2025 **IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25)**
Ad hoc review of 6 articles on Precision Health and AI.

2024 **Medicine & Science in Sports & Exercise**
Ad hoc review (via supervisor) of an article on wearable technologies for monitoring mobility disorders in people with Parkinson's disease.

Teaching

McGill University

Fall 2025 **Teaching Assistant –** EDKP 447 (Motor Control).

Fall 2024 **Teaching Assistant –** EDKP 447 (Motor Control).

University of Paris (AFREP)

Fall 2025 **Graduate Student Instructor –** UE 5.4 (Research Initiation).

Winter 2025 **Graduate Student Instructor –** UE 5.3 (Research Initiation).

Fall 2024 **Graduate Student Instructor –** UE 5.4 (Research Initiation).

Winter 2024 **Graduate Student Instructor –** UE 5.3 (Research Initiation).

Mentoring

2024 ***Dahlia Leibovich**, B.S. Student in Kinesiology, McGill U. (co-author J.04 C.01 P.12)*
Project: Exploring brain locomotor circuits using PET data.
Next: Visiting B.S. Student in Kinesiology, University of South Wales, Australia

2024 ***Laura Higuita**, MITACS B.S. Student in BioEngineering, UAO, Columbia*
Project: Detection of freezing of gait in Parkinson's disease.
Next: B.S. Student in BioEngineering, UAO, Columbia

Invited Talks

McGill University *Guest Lectures on the Neurosciences of Learning, Memory and Attention*

2025 – 2026 **CHEM 110.** General Chemistry 1.

2024 – 2025 **CHEM 120.** General Chemistry 2. (with Hilary Sweatman)

2024 – 2025 **PSYC 204.** Introduction to Psychological Statistics. (with Hilary Sweatman)

Relevant Training

2024 **Researchers and Entrepreneurs Network Program (RCE)**, V1 Studio

2023 **Integrating sex and gender into health research**, Canadian Institute of Health Research

2023 **TCPS 2 CORE-2022 (Research Ethics)**, Government of Canada

2022 **Gait Analysis & Rehabilitation using CAREN (Operator Level 1)**, Motek Medical

2021 **AFGSU 1/2 (emergency care)**, French Ministry of Health